

ML Algorithms 30/10/26

Questões

Quais os algoritmos

Isso envolve modelos de LLM ou o que?

Pq nas duas fala de dataset, me parecem mais coisa pra análise baseada em dados, modelos mais especializados talvez

Motivação

Machine learning pipeline

↳ Set of operations
to produce a model

↳ mathematical representation of a
real world-process

↳ a function that takes input
data and produces an output
(classification, sentiment, recommen-
dation or clusters)

Performance of a model

↳ by using evaluation metrics

↳ precision & recall

↳ accuracy

Include ML into software systems because it solves problems
that can be too complex to be solved traditionally
ML might be the right way to pursue it.

For instance

- Perceptive problems → large number of elements
can be fixed with with different representations
speech recognition
sentiment analysis
- Multi-parameters problems
↳ stock prices prediction

Goal → make models available to software systems

1. to provide functionalities

- Recommendation
- Top items
- Classification
- Content generation
- Spam detection
- Fraud

2. Time to deploy

Between a week to 3 months or even an year

↳ Main Challenges people face when developing ML models

↳ Scale, Version Control, Reproducibility, aligning stakeholders.

Deployment Gap

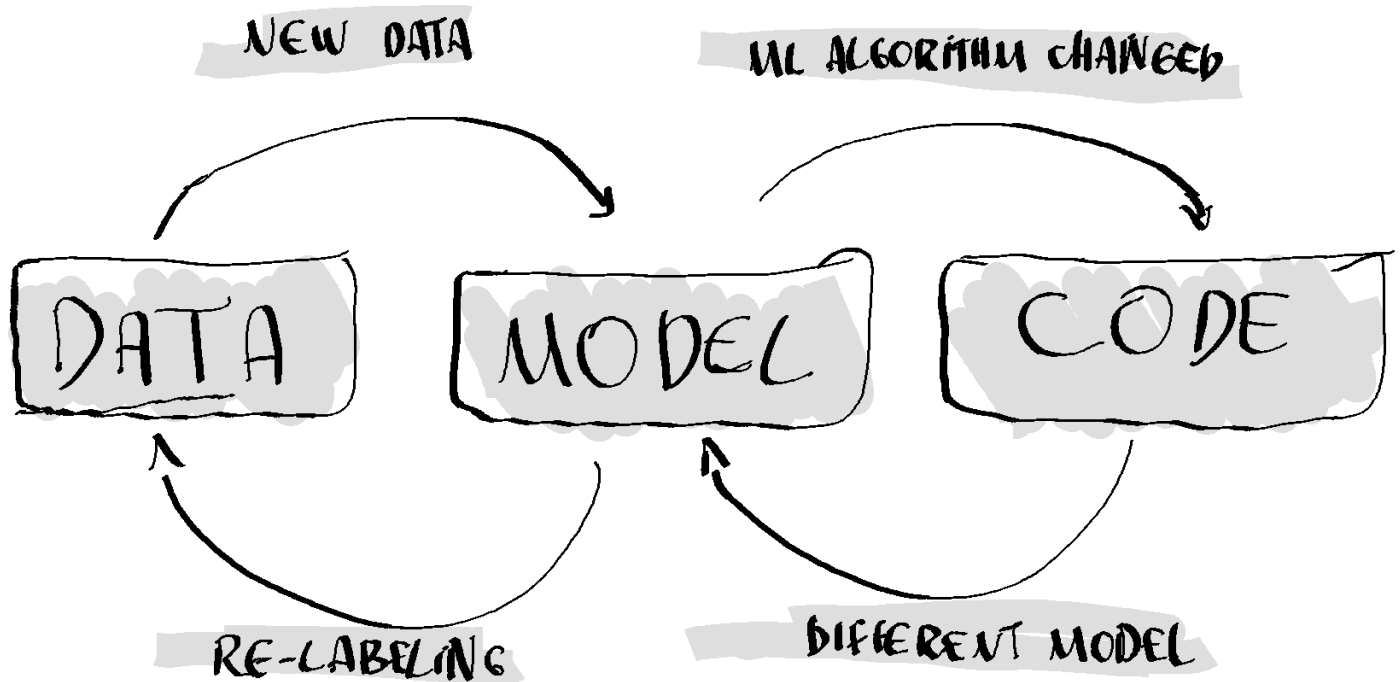
Business need it → we need to deploy it
to prod

Other software systems populate data back,
then model is trained again

Deployment is different: Data, ML model and Code

↳ build trigger → combination of code change,
data change or model change

Principle: Changing Anything Changes Everything



Examples of scenarios to change an ML App

- 1 Model starts to decay, behave abnormally
needs new data to re-train the model
 - 2 New data, re-label training data, retrain the model
 - 3 Wrong assumptions, needs to re-train the model
 - 4 Business objective change
- ML is expanding, business needs are changing fast

* Details

1. Model decay because real-time data changes
It hasn't been seen during the model training

MLOPS

- Extension of Devops Methodology to include ML and Data Science assets as first-class citizens within the Devops ecology
- Alternatively uses term MLE (Machine Learning Engineering)
 - ↳ MLE: Use of scientific principles, tools and techniques of ML and traditional software engineering to design and build complex computing systems. Encompasses all stages from data collection, model building, make it available for use by the product
- Separating ML Model development from process that delivers it lowers the quality and agility.
Usando lápis pra diferenciar, pra contrastar.
Agora é um lápis, o outro era caneta

- Evolution of #MLOPS

Tem uma imagem no Obsidian